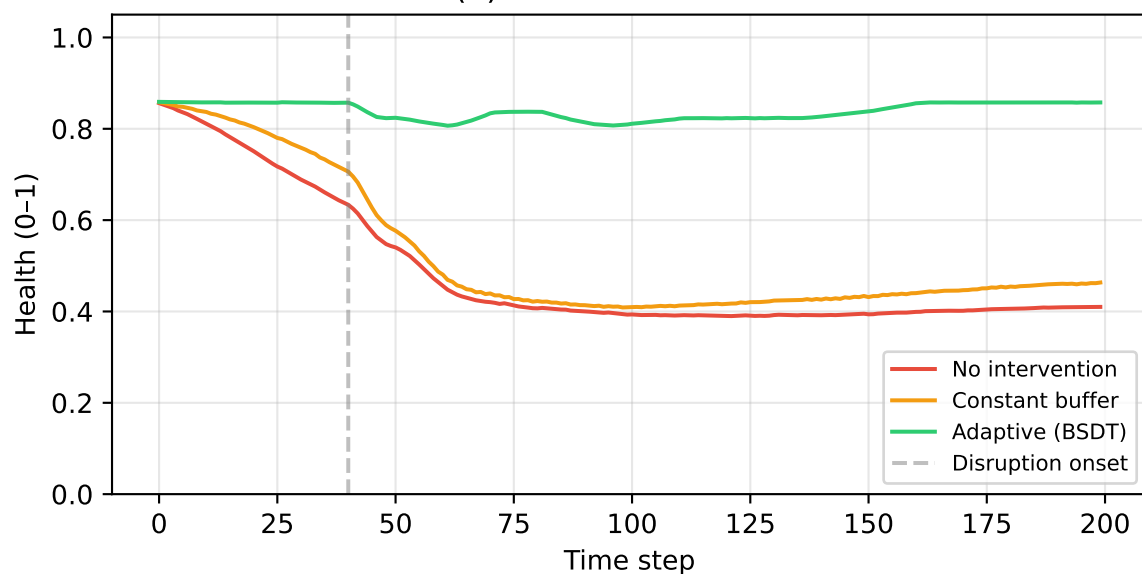
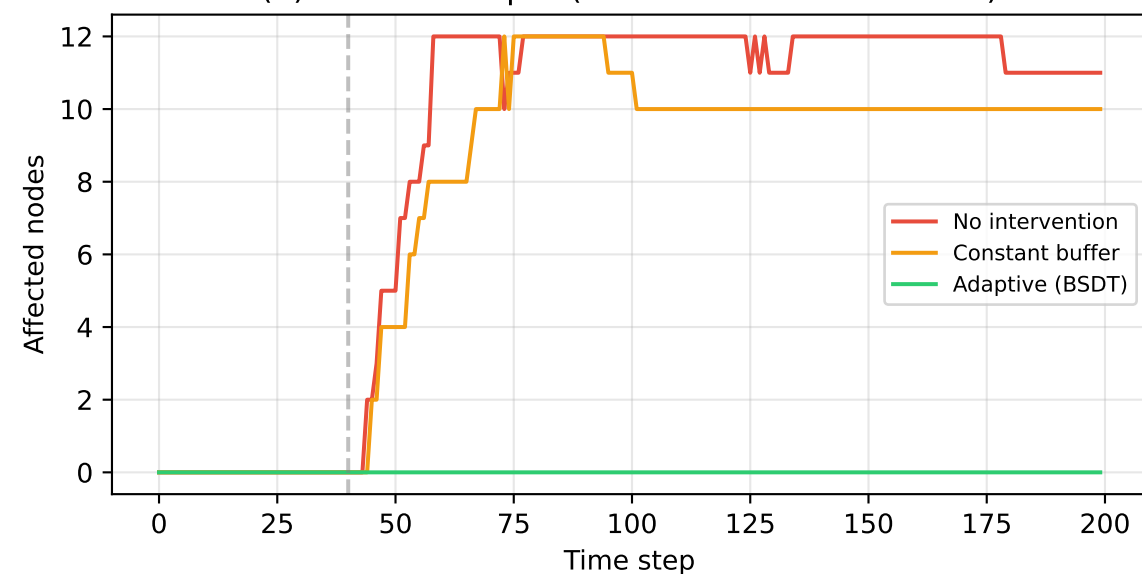


# Supply Chain BSDT — COVID-19 Semiconductor Shortage

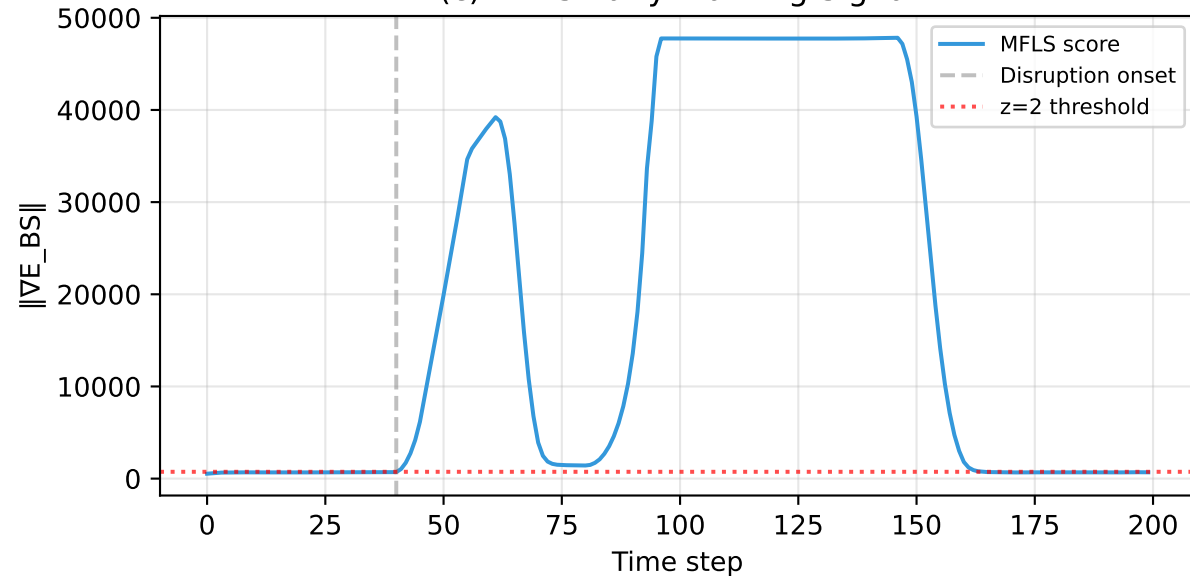
(a) Mean Node Health



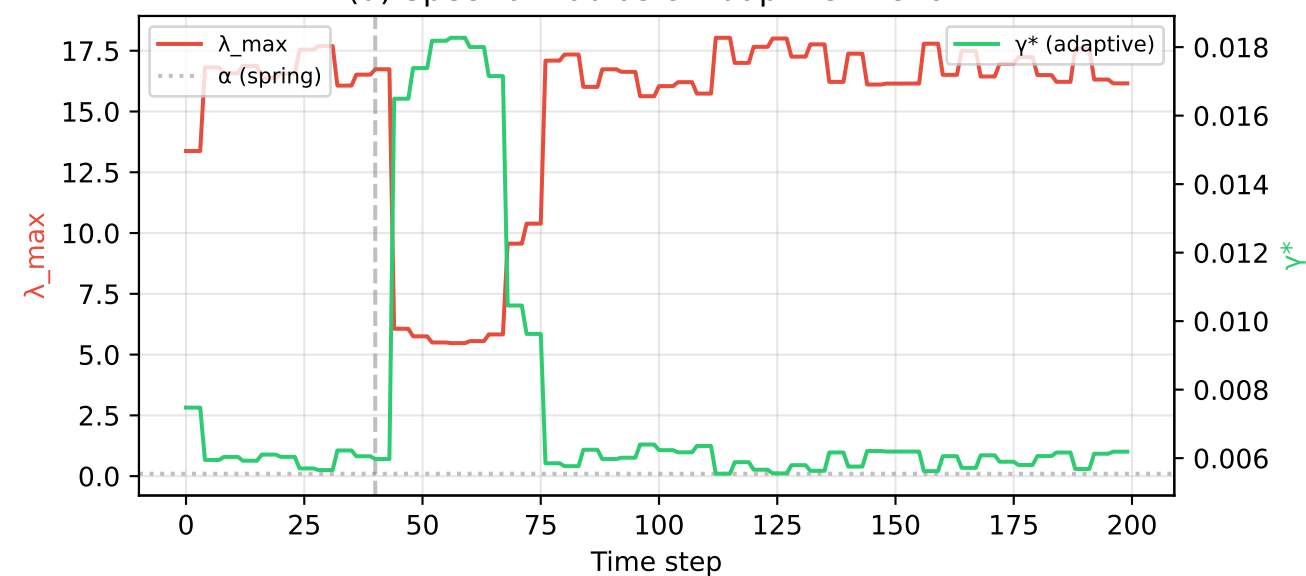
(b) Cascade Depth (nodes with health < 50%)



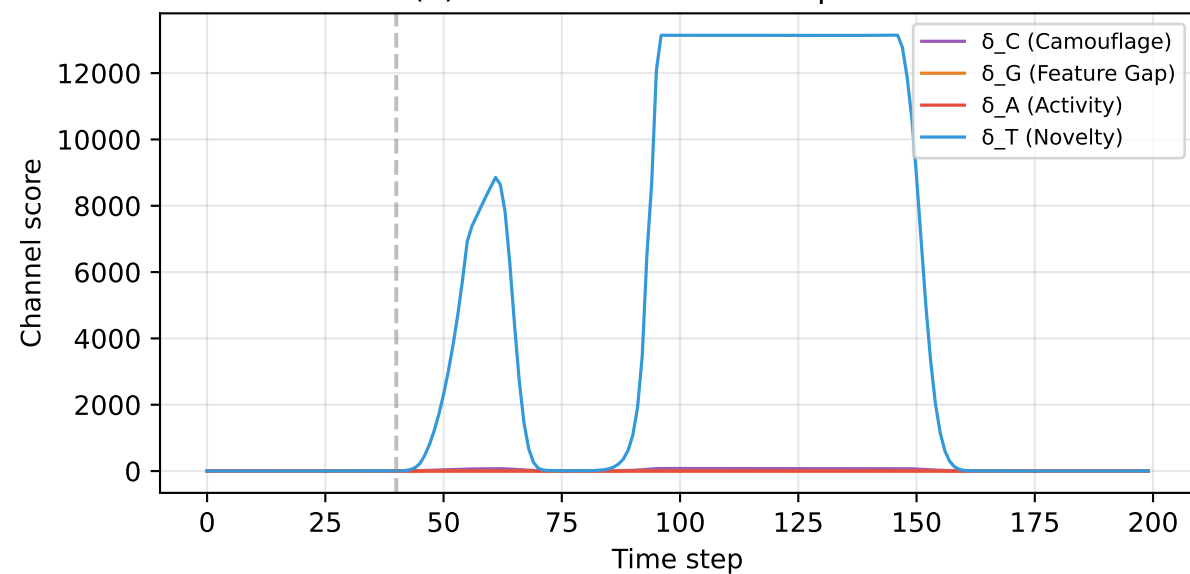
(c) MFLS Early Warning Signal



(d) Spectral Radius & Adaptive Friction



(e) BSDT Channel Decomposition



(f) Gradient Alignment ( $\nabla E_{BS}$  vs  $-\nabla \Phi$ )

